{DM - functions}

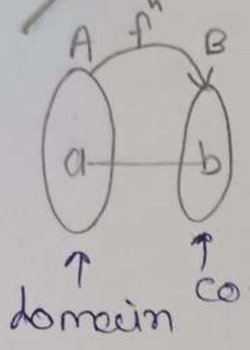


image of a = b
preimage of b = a

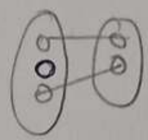
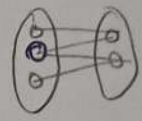
• Range ≡ Reach
↪ which ele. of co-domain are reachable from domain

$f : A \to B$ $\forall_{a \in A}$ $\exists_{b \in B}$ $f(a) = b$ ← unique

- $\forall_{a \in A}$ a has exactly one image!

$|A| = n$, $f(\text{Domain}) = \text{Range}$
$|B| = m$

$\boxed{\#\text{ total } f^n = m^n}$

 +  } NOT even a function

→ $(f+g)(x) = f(x) + g(x)$
$(fg)(x) = f(x) * g(x)$

$A \xrightarrow{R} B \xrightarrow{S} C$ (Transitivity)
$h : A \to C \Rightarrow h(a) = S(R(a))$
composition → $h = S \circ R$ ← पहला

→ one-one / Injective

$F : D \to C$ ← for each $y \in C$, at most one preimg

If $f(a) = f(b)$ then $a = b$

• $|A| \le |B|$

→ onto / surjective

$f : D \to C$ ← for each $y \in C$, at least one preimg

$\forall_{y \in C}$; $\exists_{x \in D}$ $f(x) = y$

• $|A| \ge |B|$

→ Bijective (1-1 & onto)

$f : D \to C$ ← for each $y \in C$, Exactly one preimg

$\forall_{y \in C}$; exactly one $x \in D$ $f(x) = y$

• $|A| = |B|$

→ $g = f^{-1}$ (inverse)

"Reverse the mapping" given by function f.
(g also should be function) iff f is bijective

invertible f^n == Bijective f^n

→ $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$

→ $f \circ f^{-1}(x) = x$

$F : A \rightarrow B$, $g : B \rightarrow C$

1) If F, g both 1-1 then $g \circ F$ 1-1.

2) ——— " ——— onto ——— " ——— onto.

3) ——— " ——— bijective ——— " ——— bijective

{DM - Set theory}

\ = − ≡ set diff.

$a \in S \equiv a \notin \overline{S}$

→ $A - B = A \cap \overline{B}$

→ $A = B$ means $\forall_x \; x \in A \leftrightarrow x \in B$

means $A \subseteq B$ & $B \subseteq A$

→ $|S| = n$ (all has 2 choice)

power set $P(S) = 2^n$

$P(P(S)) = 2^{(2^n)}$

→ $x \in S$ then $x \notin P - S$

$x \notin B$ then $x \notin B - S$

$x \in A - B \leftrightarrow x \in A$ and $x \notin B$

→ $(\overline{A} - B) - C =$

$(\overline{A} - C) - (B - C)$

$A \Delta B = (A - B) \cup (B - A) = A \oplus B$

→ $x \in P(A) \overset{\text{iff}}{\longleftrightarrow} x \subseteq A$

→ # subset $= 2^n$

proper subset $= 2^n - 1$

→ $A - B = \overline{B} - \overline{A}$

$= (A \cup B) - B$

→ $x \notin A - B$ means

$x \notin A$ or $x \in B$

→ Idempotent law :-

iff $x \# x = x$

→ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

→ precedence : $\overline{A} > A \cap B > A \cup B$

→ Identity law :-

$A \cup \phi = A \quad A \Delta \phi = A$

$A \cap U = A$

↳ identity ele.

Identity ele. "e" for opⁿ "#"

$x \# e = x$ and $e \# x = x$

→ $(A - B) - C \neq A - (B - C)$

→ ordered pairs

- order
- Repetition

} Matters

$(a, b) = (c, d) \overset{\text{iff}}{\longleftrightarrow}$

$a = c$ & $b = d$.

- Finite (order - n triple)

→ power set: $P(A) \cap P(B) = P(A \cap B)$

$P(A) \subseteq P(B)$ then $A \subseteq B$

→ set

- order
- Repetition

} Does NOT matter

→ sequence

- order
- Repetition

} Matters

- finite or infinite

→ $A \times B = \{(x, y) \mid x \in A, y \in B\}$

order-2 tuple.

$A \times \phi = \phi$

$A \times A = A \times A$

$A \times B \neq B \times A$; $A \neq \phi, B \neq \phi$

$|A| = m$, $|B| = n$ } $|A \times B| = m \cdot n$

$a | b \equiv$ a divides b $\equiv$ b is divided by a

2

→ $(A \times B) \times C \neq A \times (B \times C) \neq A \times B \times C$

2-tuple ← → 2-tuple; $A \times B \times C$ ↑ 3-tuple

→ $R : A \to B$, $R \subseteq A \times B$

$A \to B$, $a \to b$: a is related to b, b is NOT related to a. $\equiv aR_b$

$|A| = m$, $|B| = n$

$|A \times B| = m \cdot n$

possible Relation $= 2^{m \cdot n}$

⇝ Take care of Base set.

Ex: $Z \times N \to Z \times N : R \Rightarrow R$ on $Z \times N$

$(a,b) R (c,d)$ } $a, c \in Z$, $b, d \in N$ } MUST. then only move further

↑ Base set

→ Reflexive	→ NOT Reflexive	→ Irreflexive
$R : A \to A$ $\forall x \in A (xRx)$. other can be there.	$R : A \to A$ $\exists x \in A (x \not R x)$ ⇎	$R : A \to A$ $\forall x \in A (x \not R x)$

→ symmetric	→ NOT symmetric	→ Anti-Symm
$R : A \to A$ $\forall a, b \in A (aR_b \to bR_a)$ xRy & yRx, $x \not R y$ & $y \not R x$, xRx, $y \not R y$ } Allowed ✓	$R : A \to A$ $\exists a, b (aR_b$ & $b \not R a)$ → Asymmetric Anti-symm. & Irreflexive $\forall a, b \in A (aR_b \to b \not R a)$	$R : A \to A$ if aR_b & bR_a then $a = b$. if $a \neq b$ then aR_b & bR_a × aR_b & $b \not R a$ ✓ $a \not R b$ & $b \not R a$ ✓ aRa ✓

→ Transitive $R : A \to A$

If aR_b & bR_c then aR_c for $\forall a, b, c \in A$.

→ NOT Transitive

$\exists a, b, c \in A (aR_b$ & bR_c & $a \not R c)$

① $a \to b \to c$, $a \not\to c$

② $a \leftrightarrow b$ But $a \not R a$ or $b \not R b$

→ Equivalence Relation

iff Reflexive, symmetric, Transitive } (RST properties satisfied)

$a \equiv b \pmod{n}$ iff $rem(a, n) = rem(b, n)$ (congruence modulo m)

→ Partition

① $A_i \cap A_j = \phi$; $i \neq j$

② $A_1 \cup A_2 \cup \ldots A_n = A$

③ $A_i \neq \phi$, ④ $A_i \subseteq A$

↪ collection of disjoint set.

→ each partition ⇒ eqⁿ class

$[x]_R = \{y \mid xRy\} = \{y \mid yRx\}$; R = eqⁿ Rel.

↪ every ele. is related to everyone.

$R = (E_1 \times E_1) \cup (E_2 \times E_2)$; $E_1 \cap E_2 = \phi$

$|R| = |E_1|^2 + |E_2|^2$

→ xRy iff $[x]_R = [y]_R$ } every $a \in A$ belongs to exactly one eq. class.

$x \not R y$ iff $[x]_R \cap [y]_R = \phi$

→ R be an equivalence relation on set A
- every eq. class of R is subset of A.
- set of eq. class of R is subset of P(A).

→ **Partial order** (Poset) (POR) - Some order we want.

Ex: $(R, \leq)$ $(R, \subseteq)$ $(R, |)$ divide

- Reflexive ++
- Antisymm. ++
- Transitive.

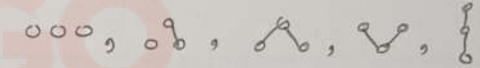
} RAT properties → xRy iff x "MUST" be done before y.

→ **Total order** (TOR) → comparable x, y iff xRy or yRx
- POR ++
- every pair of elements is comparable.

TOR $\rightleftarrows$ POR (TOR ⇒ POR; POR ⇏ TOR)

Ex: $(Z, \leq)$ ✓ $(Z^+, |)$ X

→ **Hasse Diagram for POR** :-



} unlabelled HD

Do NOT show : $aRb \Rightarrow$ b | a, $b \not R a$
- self loop
- Trans. edge

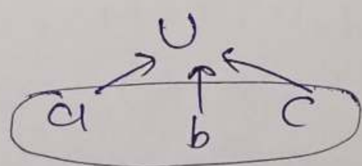
⇒ Remove "Arrows" AND all dir. upward

b ⋮ a ⇒ $aRc + cRb$ transitively.

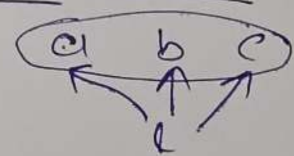
- minimal ele.
- maximal ele.
- Greatest/Maximum (G): $\forall a, aRG$
- least / Minimum (L): $\forall a, LRa$

• If exists then unique ⇐ (⇏)

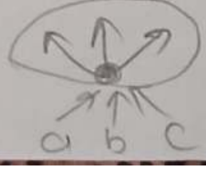
- upper bound of $\{a, b, c\}$



- lower bound of $\{a, b, c\}$

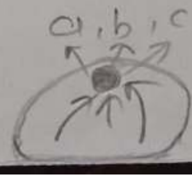


- least upper bound $\vee$: In UB(x), find least element



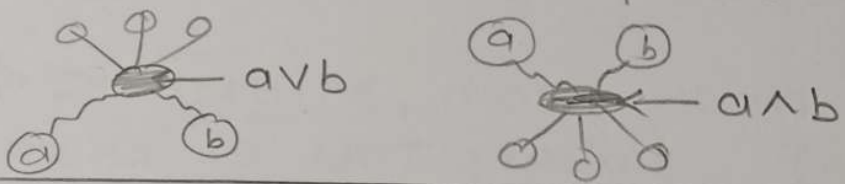
"If exists then unique"

- Greatest lower bound $\wedge$: In LB(x), find greatest element.



→ IF Poset has :
more than one maximal ele. → NO Greatest
more than one minimal ele. → NO least.

1) $a \vee a = a$, $a \wedge a = a$; $\vee$: LUB & $\wedge$: GLB
2) If aRb then $a \wedge b = a$, $a \vee b = b$; b | a (Comparable)
} for 2 ele.
3) If a, b are NOT Comparable :



; unique, if exist (NOT unique then DNE)

→ Hasse Dia. for TOR ⇒ chain / straight line

→ **Lattice** :- Poset in which every pair of ele. have GLB, LUB.
→ check only Incomparable pairs.

→ on 'n' ele., the largest Possible POR is a chain (TOR)

→ "A × A" is always Eq. Relation for A ; SO NEVER POR.

→ every Relation on 'ϕ' :-
- Ref ✓ Irref ✓
- Sym ✓ Antisym ✓
- Trans ✓ Asym ✓

'A × A' : full Relation
→ If full Relation is POR then $|A| = 1$.

→ |largest possible POR| $= \frac{n(n+1)}{2}$ ⇒ TOR

|smallest Possible POR| $= n$ ⇒ Identity Rel aRa ∀a∈A.

GLB = GCD } when
LUB = LCM } R(N, |)

→ $a \vee (a \wedge b) = a$
$a \wedge (a \vee b) = a$
for lattice

; ∀a, $a \wedge e' = a$
$a \vee e = a$
↑ Identity ele.

→ IF (A, R) is poset then
$\forall B \subseteq A$ (B, R) is also Poset.

→ sub-lattice (S) :
S of Lattice L
- ① subset of L
- ② S ⇒ lattice
- ③ GLB, LUB for any a, b ∈ S must be same in L, S.

→ taking a, b then take their GLB, LUB also in sub-lattice S.

→ To check if S is sublattice of L : just check ∨, ∧ of Non-comparable pairs of S ; must same as L.

→ In "finite" Poset:

Unique maximal $\overset{\text{then}}{=}$ Greatest

Unique minimal = least

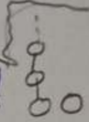
$(N, \le)$: maximal = DNE

$(Z, \le)$: maximal = minimal = DNE

S

→ No maximal ele. ↓↑ Poset is infinite

→ In "Infinite" poset



Unique maximal ↛ Greatest

Unique minimal ↛ least

→ In lattice, more than one maximal/minimal element is NOT possible. (either unique or DNE)

(finite lattice) (infinite)

→ lattice L;

$\forall_{x,y \in L} \left(xR(x \vee y) \;\&\; (x \wedge y)Rx \;\&\; (x \wedge y)R(x \vee y) \right)$

$xRy \xleftrightarrow{\text{iff}} x \wedge y = x$

$x \vee y = y$

in lattice

- commutative
- associative
- absorption
- idempotent

↑ $(\wedge, \vee)$

Properties satisfied by every lattice

→ Types of lattice :-

1) Bounded → Identity prop.
2) Complemented → Compl. prop
3) distributive → Distributive prop.
4) boolean → all 3 above.

→ every finite lattice is bounded (∵ unique greatest/least) G L

$\forall x ;\; x \vee G = G$

$G \wedge x = x$

$x \wedge L = L$

$L \vee x = x$

Ex: $(N, \le)$, $(\omega, 1)$

→ Identity : $a \# e = a = e \# a \;;\; \forall a$

G for GLB ∵ $x \wedge G = x$

L for LUB ∵ $x \vee L = x$

→ Dominator :

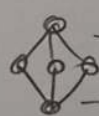
G for LUB ∵ $G \vee x = G$

L for GLB ∵ $L \wedge x = L$

→ Distributive :

$\forall x,y,z \quad x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$

$x \wedge (y \vee z) = (x \wedge y) \vee (y \wedge z)$



&

↳ NOT distributive

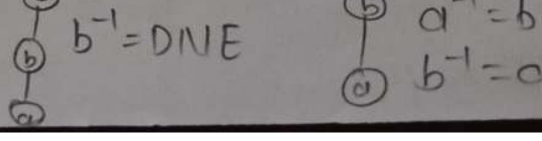
→ Complement :

In bounded

iff $a \vee b = G$, $a \wedge b = L$ then a is complement of b

$\forall a, b \in A$

→ Toset be Complemented lattice iff ≤2 elements.



→ L is distributive iff there is NO sublattice of L which is kite or pentagon.

∴ ≤ 4 ele then distr. always.

→ Distributive lattice $\xrightarrow{}$ at most one complement for every ele.
Distributive lattice $\not\leftarrow$ at most one complement for every ele.

→ If L is Total order → Distributive

a|b iff $\exists_{int\ q}$, $b = q \cdot a$
(N, |) (ω, |) : poset
(Z, |) : NOT poset
(∵ $-1 | 1$ & $1 | -1$)

→ $\forall a \in \underline{D_n}$ $a|n$. (set of all divisor of n)

(Dn, |) — poset
— least : 1
— greatest : n
— lattice
LUB = LCM
GLB = GCD
— Bounded
— Distributive

Ex: $D_{16} = \{1, 2, 4, 8, 16\}$

→ n is square free iff n is NOT divisible by p^2 where p = prime.
(so do prime factorization and check)

∴ (Dn, |) is Complemented / iff n is square-free
(Boolean lattice)

$n = P_1 \times P_2 \times P_3 \times P_4$

$(P_1)^{-1} = \frac{n}{P_1}$

$(P_2 \times P_3)^{-1} = \frac{n}{P_2 \times P_3}$

} Complements in lattice.

Refinement of partition
≡
further fragmentation

— Reflexive
— Anti-sym
— Transitive

(Π, Refine) — poset
on set S.
— lattice
— Bounded
— Greatest : {S}
— least : finest partition
— may not be distributive
— may not be boolean Alg.

→ DFA is refinement of mDFA.

→ every eq. rel. on set A is refinement of A×A.

→ Topological order in Poset Hasse dia.
‖
Compatible Total Order

{DM - Group theory}

→ Algebraic structure = (A non-empty set + one or more closed op^n (n-ary))

Ex: $(N, +, \times)$, $(Z, +)$
(N, F) ; $F(a,b,c) = ab + c$

→ Algebraic structure ≡ Abstract structures (here with single binary op^n) ≡ Group

• closure property :- set S is closed under op^n # iff $\forall a,b \in S$ $a \# b \in S$

↳ Binary operation must be closed.

• Associative property :- $\forall a,b,c \in A$, $(a\#b)\#c = a\#(b\#c)$

• Commutative property :- $\forall a,b \in A$ $a\#b = b\#a$

• Identity Property :- $\exists e \in A$, $a\#e = e\#a = a$; for $\forall a \in G$.

↳ for every closed structure, at most one identity element.

• Inverse Property :- $\exists e \in A$, $a\#b = e = b\#a$ then $b = a^{-1}$ for $\forall a,b \in S$ $(\& a = b^{-1})$

→ for identity ele. 'e' ; $e^{-1} = e$

---

i) Magma (groupoid) : closure

ii) semigroup : closure + associativity

iii) Monoid $(S, \#, e)$: closure + associativity + identity

iv) Group : Monoid + inverse

v) Abelian : Group + commutative

(Abelian monoid) : monoid + Commutative

$* : S \times S \rightarrow S$

functions = # operation

$= |S|^{(|S| \times |S|)}$

Commutative binary op. $= n^{\frac{n^2+n}{2}}$; $|S| = n$

→ set of n^{th} root of unit is "abelian group".

→ Addition modulo-n group $(Z_n, \oplus_n)$ is "abelian group"
$(a+b) \% n = \oplus_n$

→ for every group; 'unique' identity ele.
so, every element has 'unique' inverse.

→ Group has both left & right cancellation property.

→ for any group, $(a \# b)^{-1} = b^{-1} \# a^{-1}$

→ In Group $(G, *)$ $ax = b$ has unique solⁿ
$x = a^{-1} b$

→ $a * b = e$, $b * a = e$ → does not mean that, it is Commutative.
$a * b \neq b * a$

→ cayley table (for group)
i) only created for finite group
ii) Row & colⁿ of identity ele. is Exactly same as header
iii) each ele. $g \in G$ appears exactly once in each row & in each colⁿ.

→ To check associativity, Never include Identity element. (bcoz it always satisfied)

→ In order-4 group templates, only 2 Non-isomorphic groups.

→ $a^m \cdot a^n = a^{m+n}$
$a^n \cdot a^{-n} = e$
$(a^m)^n = a^{mn}$

} hold on every group, because of associativity and inverse.

→ **subgroup** :- subset which is a group

subgroup $H = \{e, a_1, a_1^{-1}, a_1^2, ---\}$; $G = (\{e, a_1, a_2 --\}, *)$

if we take 'a_1' then we have to take $a_1^{-1}, a_1^2, a_1^3, ---$ also ($H = \langle G \rangle$)

→ coprime ≡ Relatively prime no. a & b
$GCD(a, b) = 1$.

→ If p is prime number then $1, 2, 3 ---, p-1$ is coprime with p.

→ $(Z_n, \circledast_n)$: NOT group (bcoz, inverse property is not satisfied.

– To make 'group', remove elements which has NOT inverse.

∴ U_n :- elements having inverse.
$\forall a \in U_n$ (a coprime with n)
$GCD(a, n) = 1$

→ U_8 & U_{12} are Isomorphic to each other

→ $\dfrac{|G|}{|S|}$; $S \subseteq G$ (finite)
$|G|$ ← order of a group
$|S|$ ← order of an ~~ele~~ subgroup.

→ $\dfrac{|G|}{|x|}$; $x \in G$ finite
$|G|$ ← order of a group
$|x|$ ← order of an element.

→ order of an element $a \in Z_n$ (under $\oplus_n$) is $\dfrac{n}{gcd(a, n)}$

→ Group of order ≤ 4 is Abelian
→ every group of prime order is Abelian
→ Smallest Non-Abelian group has order = 6.
→ $a^0 = e$
$a^n * a^{-n} = e$
→ order of an ele ;
smallest +ve integer n, such that $g^n = e$
→ order of 'e' is always 1.
→ In ∞ group, order of an ele. may or may NOT be finite.
→ generator, $g \in G$; if $\forall a \in G$ $g^n = a$
→ cyclic group :- at least one generator
→ Identity ele. only generates itself.
→ smallest Non-cyclic group :- order-4
→ If g is generator then g^{-1} is also a generator.
→ order of generator = order of the cyclic group.
→ every cyclic group is Abelian.
→ subgroup $H \subseteq G$
1. $e \in H$
2. $x, y \in H \Rightarrow x * y \in H$
3. $x \in H \Rightarrow x^{-1} \in H$

→ subgroup H_1 & H_2
$H_1 \cap H_2$ also subgroup
$H_1 \cup H_2$ may or may NOT be subgroup

→ smallest subgroup contains 'a' = cyclic subgroup generated by 'a'

{DM-Graph theory}

→ 1 — 2 $\{1, 2\}$ ← set (unordered)

1 → 2 $(1, 2)$ ← ordered pair

→ multigraph :- undirected graph & NO self loop

Directed multigraph :- Any directed graph

pseudo graph :- Any undirected graph

simple graph :- undirected, NO self-loop, NO multiedge

→ In a Multi graph; we have multiset. (duplicates)

→ self loop gives a degree of **2**.

→ Isolated vertex ⇒ vertex with degree-0

Pendant vertex ⇒ vertex with degree-1

→ order(G) = |V|

size(G) = |E|

→ degree sequence (n-term) = usually, descending order of vertex degree.

→ In undirected graph, we can NOT create odd no. of odd-degree vertices. $\sum_v \deg(v) = 2|E|$

→ In directed graph,

Total InDeg. = Total OutDeg. = |E|

odd deg ⇒ even no.

even deg ⇒ odd or even no vertices

NOW, SIMPLE GRAPH}

Walk : vertex & edge Repetition

Trail : Walk ; No edge Repetition ⇒ closed Trail ⇒ circuit

Path : Trail ; NO vertex Repetition ⇒ cycle ⇒ path but $v_1 = v_n$

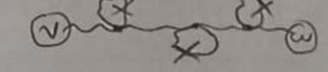
(Nothing repeat)

→ path is NOT a cycle, cycle is NOT a path

→ NO vertex Repetition $\xrightarrow{\text{implies}}$ No Edge Repetition

NO vertex Repetition $\not\longleftarrow$ No Edge Repetition

→ In a graph, If there is a walk from v to w then there is a path from v to w.



→ Distance : length of shortest path Dist(a, b)

→ Diameter : longest shortest path Dia(G)

→ Avg. degree = Ω

Min. degree = δ

Max. degree = Δ

$n \cdot \Omega = 2 \cdot |E|$

$n \cdot \delta \le 2 \cdot |E|$

$n \cdot \Delta \ge 2 \cdot |E|$

→ spanning subgraph :- vertex deletion NOT allowed

Induced subgraph :- edge deletion NOT allowed. (vertex & its incident edges can be deleted)

→ Graph Induced by S :- $S \subseteq V$

$H(S, E') \Rightarrow E' = \{ e \mid e \in E$ and both endpoints of e are in S $\}$

Graphs $= 2^{^nC_2}$

(each pair has 2 choice : put edge or NOT)

→ Order of Graph (to check) Invariant:

1) $|V|$
2) $|E|$
3) connectedness
4) Degree seq.
5) No. of 3-length cycle
6) " " 4-length "
7) " " 5-length "
8) Diameter

→ Isomorphism : $G(V_1, E_1)$, $H(V_2, E_2)$

$G \equiv H$ iff

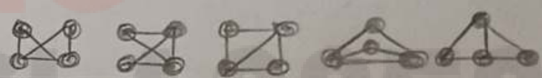
$\exists$ bijection $f : V_1 \rightarrow V_2$

s.t. $\forall a, b \; (a,b) \in E_1$ iff $(f(a), f(b)) \in E_2$

(edge preserves)

→ some Graph Invariants is NOT same → NOT Isomorphic

→ ALL Graph Invariants are same ↛ Isomorphic



→ Isomorphic is equivalence relation. (Reflexive, symm., Trans.)

→ Complement of graph

$\overline{G}(V, \overline{E})$; $\overline{E} = E(K_n) - E$

$e' = {}^nC_2 - e$

→ Compl. of Disconn. is always conn.

→ self-Complementary

$G \equiv \overline{G}$ (Isomorphic)

$e' = e \Rightarrow e' + e = {}^nC_2$; e = #edge in G.

→ $n(n-1) = 4 \cdot e$

→ Relation R on V ; $v_1 R v_2$ iff $\exists$ path b/w v_1 & v_2 is equivalence Relation.

#eq. class = # of Component.

Connected iff $\exists$ path b/w all vertices

→ simple undirected graph :- # Max. edges $= {}^nC_2$

simple Directed graph :- # Max. edges $= n(n-1)$

→ In Bipartite graph, partition can be empty.

→ bipartite ⟷ all cycle has EVEN length

bipartite ⟷ NO odd length cycle

→ $\sum_{v \in X} deg(v) = \sum_{v \in Y} deg(v)$ (Partition)

→ Acyclic graph ⇒ every pair of vertices have at most one path b/w them.

→ Tree $\xrightleftharpoons[\not]{}$ $|E| = |V| - 1$ { + ... also. }

|||

Maximally acyclic Graph on n vertices $\equiv$ minimally connected graph (with min. # of edges)

→ In tree, # of degree (1) vertices is at least two ; $n \geq 2$.

→ **Rooted Tree** :- DAG with a Root AND unique path from root to every other vertex.

Forest with Max. number of edges $\equiv$ Tree ⟶ ($|E| = n - 1$)

Tree + one edge = cycle with min. # edge

G has No cycle $\equiv$ G is forest (NOT tree directly)

→ In FBT ;

# of Internal Nodes $= L - 1$ (# leaves)

→ G with $> {}^{n-1}C_2$ edges then G is definitely "connected".

→ clique no. $\overset{\omega}{=}$ size of a Max. clique ⇒ adj. to each other

Independence no. $\overset{\alpha}{=}$ size of a Max. Ind. set ⇒ Non-adj vertices

→ S is Max. Ind. set in G ⇄ S is max. clique (compl. sub-graph) in $\bar{G}$

→ Vertex cover ⇒ which cover all edges

Edge cover ⇒ which cover all vertices

→ Vertex covering no (β) = size of min. vertex cover (MVC)

Edge covering no. (β') = size of min. edge cover (MEC).

→ $\beta' \geq \lceil \frac{n}{2} \rceil$ ∵ one edge give exactly two vertices

→ Minimal edge cover of any graph contains NO paths of length-3 & NO triangle.

→ If $S \subseteq V$ is vertex cover then $(V - S)$ is Independent set.

→ S = largest IS then $V - S$ = Min VC

→ $\alpha \geq n - \beta$ | → $\alpha \leq \beta'$

$\boxed{\alpha' \leq \lfloor \frac{n}{2} \rfloor}$

→ Matching ⇒ pairwise Non-adj. edges

→ Matching no. $(\alpha') \overset{\alpha'}{\Rightarrow}$ size of largest matching set

→ perfect matching (PM) ⇒ matching which cover all the vertices.

→ PM exists $\xrightleftharpoons[\not]{}$ $|V|$ even

$\alpha + \beta = n = |V|$

$\alpha' + \beta' = n$

→ every PM $\xrightleftharpoons[\not]{}$ Maximum matching $\xrightleftharpoons[\not]{}$ maximal matching

→ |Matching| ≤ |vertex cover| ; $\mu \leq \beta \leq 2 \cdot \mu$

→ K-colorable ⇒ using at most K-colors

→ chromatic no. (χ) ⇒ min. no. of colors needed
($\chi \leq K$)

→ $\chi \leq \Delta + 1$
(Δ = Max. deg.)

→ $\chi \geq \omega$
(ω = clique no.)

→ odd cycle ; $\chi \geq 3$

→ $\chi = m$; $|E| \geq {}^{m}C_2$

→ $\alpha \geq \frac{|V|}{\chi}$
(α = ind. set) ($\frac{|V|}{\chi}$ = avg. size of color class)

→ for Trees, $\sum$ deg. seq. $= 2(n-1)$
(for tree realization)

→ chromatic Index (χ') ⇒ edge chromatic number.

→ $\chi' = \Delta$ (class-1) or $\Delta + 1$ (class-2)

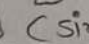
→ < 3 vertices of Max. deg (Δ) → class-1 graph $\chi' = \Delta$ (converse ✗)

No odd cycle → bipartite → $\chi' = \Delta$

Reg. graph with odd vertices → $\chi' = \Delta + 1$

→ Realization of simple Graph :-

1) # of odd deg vertices = even
2) $\Delta \leq n-1$, $\delta \geq 0$
3) At least two vertices MUST have same degree.
4) deg. seq = $d_1, d_2 \ldots d_n$
$d_1 \leq n-1$ & $d_2 - 1, d_3 - 1, \ldots, d_{d_1+1} - 1, d_{d_1+2}, \ldots d_n$ check recursively

→ cut vertex = Articulation point (single vertex) 
cut edge = Bridge (Non-cycle edge)  (single edge)

→ G → # component = K
G' = G − v → # Component = m
} $K-1 \leq m \leq n-1$; $\delta = 0$ (disconn. graph)
$K \leq m \leq n-1$; $\delta = 1$ (conn. graph)

G' = G − e → # component = n } $n = K+1$ or K

→ If every edge of graph G is a cut-edge then G is Forest (NOT tree).

→ vertex cut ⇒ set of vertices (either disconn. graph or single vertex remain)

→ connectivity number ⇒ size of smallest vertex cut
(K)

K(disconn. graph) = $K(n=1) = 0$

→ k-connected graph ⇒ $k \leq K$; $K \leq \delta$ (δ = min. deg)

→ Pendant vertex can NEVER be a cut-vertex.

→ Edge cut ⇒ set of edges

→ Edge connectivity number ⇒ size of smallest edge cut
(λ) (min-cut)

→ $\frac{n \cdot k}{2} \le e$, $\frac{nk}{2} \ge e$

→ strongly connected ⇒ $\forall_{a,b}$ path from a to b && b to a.

→ strongly connected Component (SCC) should be maximal.

→ Directed graph G + one more edge then $1 \le \#SCC \le K$

→ strongly conn. $\xrightarrow{\quad}$ / $\not\leftarrow$ weakly conn. (ignore direction)

→ Euler circuit :- 'circuit' containing "Every edge". (exactly once)

→ Euler Graph iff Euler circuit exist.

|||

connected even deg. graph.

→ Euler path

|||

Connected & Exactly 2 vertices of odd deg.

→ Directed ;

Euler circuit iff strongly comm. + Indeg = outdeg for all vertex

Euler path iff weakely comm. + outdeg = Indeg + 1 for one vertex

outdeg = Indeg − 1 for one vertex

outdeg = Indeg for remaining vertex

→ Hamiltonian cycle :- visit every vertex exactly once.

→ Hamiltonian cycle exist $\xrightarrow{\quad}$ / $\not\leftarrow$ Hamiltonian path exist

→ Euler circuit (all even deg.) $\not\rightarrow$ / $\not\leftarrow$ Euler path (exactly 2 odd deg.)

→ $n \ge 3$ & $\delta \ge \frac{n}{2}$ $\xrightarrow{\quad}$ / $\not\leftarrow$ Hamiltonian cycle.

→ If cut-vertex/cut-edge exist then NO Hamiltonian cycle exist.

→ G is Planar Graph iff there exists a planar representation of G.

→ Degree of face :- # of sides of edges it touches

$\sum_{\forall i} deg(F_i) = 2 \cdot |E|$ (in planar graph)

→ In any connected planar graph with at least 3 vertices ; Degree of face is ≥ 3.

→ G : Planar graph

G − one edge ⇒ 1) either F decrease by 1 ; if E is part of cycle
or C (#comp.) increase by 1 ; if E is NOT part of any cycle
2) |E| decrease by 1
3) |V| unchanged

For planar graph

$\boxed{V + F = E + C + 1}$

→ IF $e \nleq 3V-6$ ⟶ Graph is NOT planar.

→ $F \leq \frac{2e}{3}$ ⟶ conn. planar graph with $|V| \geq 3$ $\xrightarrow{} e \leq 3V-6$ ($\not\leftarrow$)

∵ $3F \leq \sum_{\forall i} deg(f_i)$

&& NO triangle $\xrightarrow{} e \leq 2V-4$ ($\not\leftarrow$)

∵ $4F \leq \sum_{\forall i} deg(f_i)$

→ every planar graph is 4-colorable ⇒ $\chi \leq 4$

→ entry i, j in A^K gives the no. of walks from i to j of length-K. (exactly)

→ $Trace(M^2)$ = Total deg. = 2|E|

if self loops then Total deg. = $Trace(M + M^2)$

→ $(I_n + A)^{n-1}$ has NO 0's ⟷ connected ; undir.

$(I_n + A)^n$ has NO 0's ⟷ strongly connected ; directed

→ # Triangles = $\frac{Trace(M^3)}{6}$; undirected

$= \frac{Trace(M^3)}{3}$; directed

$(M + I_n)^n \neq M^n + I_n$

α : Independence no.

β : vertex covering no (size of MVC)

β' : edge covering no. (size of MEC)

χ : chromatic no

χ' : edge chromatic no.

κ : Conn. no. (size of smallest vertex cut)

k :

ω : clique no.

α' / e_1 : size of largest matching set

▷ Regular graph (d-regular)

→ deg. seq :- $\underbrace{d, d, --- d}_{n\text{-times}}$

→ $\Delta = d$
$\delta = d$
Avg. deg = d
Total deg = $n \cdot d$

→ $|E| = \frac{n \cdot d}{2}$

→ if $|V|$ = odd then $\chi' = \Delta + 1$

→ connected or disconnected

→ diameter can be finite or ∞.

▷ Complete graph (K_n)

→ deg. of every vertex = $n-1$

→ $\Delta = \delta$ = Avg. deg. = $n-1$

→ $(n-1)$-Regular

→ deg. seq :- $\underbrace{n-1, n-1, ---, n-1}_{n\text{-times}}$

→ total deg = $n(n-1)$

→ $|E| = \frac{n(n-1)}{2}$

→ Diameter = 1

→ MUST be connected

→ $\alpha = 1$
→ $\omega = n$
→ $\chi = n$
→ $\kappa = n-1$

→ $\beta = n-1$
→ $\beta' = \lceil \frac{n}{2} \rceil$
→ $\chi' = \Delta$; n-even
$= \Delta + 1$; n-odd
→ bipartite

→ perfect matching exist when n is even.

$\alpha_1 = \lfloor \frac{n}{2} \rfloor$

→ For $n \geq 3$; K_n is hamiltonian

▷ Empty/Null/Edgeless graph (E_n)

→ deg. seq : $\underbrace{0, 0, --- 0}_{n\text{-times}}$

→ 0-Regular

→ $\delta = \Delta$ = Avg. deg = total deg = 0

→ $\beta = 0$

→ β' = does not exist

→ $\chi = 1$

→ $|E| = 0$

→ connected iff $n = 1$ o/w disconnected

→ Diameter = 0 when n = 1 else Diameter = ∞

→ Complete bipartite

▷ Path graph (P_n)

→ Deg. seq. = $\begin{cases} 0 & ; P_1 \\ 1, 1 & ; P_2 \\ \underbrace{2, 2 -- 2}_{n-2 \text{ times}}, 1, 1 & ; P_{n \geq 3} \end{cases}$

→ Diameter = $n-1$

→ $\chi = 2$

→ bipartite

→ $|E| = n-1$

→ Total deg = $2(n-1)$

→ connected

→ $\alpha = \lceil \frac{n}{2} \rceil$

→ $\omega = \begin{cases} 2 & ; n \geq 2 \\ 1 & ; n = 1 \end{cases}$

$n \geq 2$

→ $\beta = \lfloor \frac{n}{2} \rfloor$

→ $\beta' = \lceil \frac{n}{2} \rceil$

▷ cycle graph (C_n) ; $n \geq 3$

→ Deg. seq : $\underbrace{2, 2, ---2}_{n\text{-times}}$

→ 2-regular

→ $\delta = \Delta$ = Avg. deg = 2

→ total deg = $2 \cdot n$

→ perfect matching exist when n is even.
$e_1 = \frac{n}{2}$

→ C_n is Hamiltonian

→ $|E| = n$

→ Diameter = $\lfloor \frac{n}{2} \rfloor$

→ $\alpha = \lfloor \frac{n}{2} \rfloor$

→ $\beta = \lceil \frac{n}{2} \rceil$

→ $\omega = \begin{cases} 2 ; n \geq 4 \\ 3 ; n = 3 \end{cases}$

→ $\beta' = \lceil \frac{n}{2} \rceil$

→ $\chi = \begin{cases} 2 ; n = \text{even} \\ 3 ; n = \text{odd} \end{cases}$

→ $\kappa = 2$

→ bipartite

▷ wheel graph (W_n) ; $|V| = n+1$

$\boxed{W_3 = K_4}$

→ Deg. seq : $n, \underbrace{3, 3, ---3}_{n\text{-times}}$

→ $\delta = 3$

→ $\Delta = n$

→ Total deg = $4n$

→ Avg. deg = $\frac{4n}{n+1}$

→ $\kappa = 3$

→ $\alpha = \lfloor \frac{n}{2} \rfloor$

→ $\omega = 3$

→ $|E| = 2n$

→ Diameter = 2

→ $\beta = \lceil \frac{n}{2} \rceil + 1$; $C_n + 1$

→ $\beta' = \lceil \frac{n+1}{2} \rceil$

→ $\chi = \begin{cases} 3 ; n = \text{even} \\ 4 ; n = \text{odd} \end{cases}$

▷ HyperCube graph (Q_n)

→ $UV \in E$ iff Hamming dist $(U, V) = 1$

→ $|V| = 2^n$

→ n-regular

→ deg. seq : $\underbrace{n, n, ---n}_{2^n \text{ times}}$

→ $\delta = \Delta$ = Avg deg = n

→ total deg = $n * 2^n$

→ $|E| = n * 2^{n-1}$

→ Diameter = n

→ $\alpha = 2^{n-1}$

→ $\chi = \begin{cases} 2 ; n \geq 1 \\ 1 ; n = 0 \end{cases}$

→ bipartite

▷ star graph

→ $\alpha = n - 1$

→ $\omega = 2$

→ $\chi = 2$

→ $K_{1,n}$ (bipartite)

▷ Complete bipartite graph ($K_{m,n}$)

→ $\forall a \in X, \forall b \in Y, ab \in E$

→ $|V| = m + n$

→ $|E| = m \cdot n$

→ Diameter = $\begin{cases} 1 ; K_{1,1} \\ 2 ; K_{m,n} \quad m, n > 1 \end{cases}$

→ Deg. seq : $\underbrace{n, n, -n}_{m\text{-times}}, \underbrace{m, m, --m}_{n\text{-times}}$

→ $\chi = 2$; $m, n \geq 1$

$\alpha = \max(m, n)$

$e_1 = \min(m, n) = \kappa$

→ $\Delta = \max(m, n)$

→ $\delta = \min(m, n)$

PL

$\neg > \wedge > \vee > \rightarrow > \leftrightarrow$: precedence

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– Team GO Classes